

WELCOME

Hello Everyone I'm

Rung Pyae Sone (Marco)

3rd year Mechatronics Engineering student, studying at AGH
University of Science and Technology, in Krakow, Poland.

1

Internship

9+

Months Experience

250+

Lab Trials Analyzed

7+

Projects

Hire me →

Download CV



Edit with  Lovable

Thesis Project: Upcoming



DESCRIPTION

This thesis focusses on the design and implementation of a vision-based object tracking system using a 2-DOF pan-tilt mechatronic platform. The system utilizes a camera connected to a processing unit to detect and track a moving object (ideally aerial object such as drone) in real time.

DEVELOPMENT STAGES

- 1 Implementing a basic tracking system capable of detecting and following a moving object.
- 2 Closed-loop control using tracking error, along with PID control to improve stability, accuracy, and response time.
- 3 Predictive algorithms such as the Kalman Filter to estimate the future position of the moving object.

SYSTEM COMPONENTS & TOOLS

Mechanical System	
> Mechanism	2-DOF Pan-Tilt
Electronics Hardware	
> Control Unit	Arduino Uno
> Processing Unit	Raspberry Pi 5 (8GB RAM)
> Vision System	Raspberry Pi Camera Module v2

Electronics Hardware

- > Control Unit
- > Processing Unit
- > Vision System
- > Actuators
- > Indicator

Arduino Uno

Raspberry Pi 5 (8GB RAM)

Raspberry Pi Camera Module v2

MG996R Servo Motors

KY-008 Laser Sensor Module

Software & Control

- > Arduino
- > Raspberry Pi
- > Analysis Tools

C/C++

Python

MATLAB, Simulink

HARDWARE & SENSORS

Thesis Project



KY-008 Laser



Feetech ST3215 Servo



Raspberry Pi
Camera v2

HARDWARE & SENSORS

Thesis Project



KY-008 Laser



Feetech ST3215 Servo



Raspberry Pi
Camera v2



Arduino Uno R4



Pan-Tilt



Raspberry Pi 5 (8Gb Ram)

SYSTEM ARCHITECTURE

```
Camera → Raspberry Pi → (Vision Processing & Prediction)
                        ↓ Serial Communication ↓
Arduino → Servo Motors → Pan-Tilt Motion
```

Celestial Navigation System



DESCRIPTION

The Celestial Navigation System is a mechatronic platform designed to determine orientation by tracking celestial objects in real time. It uses a 2-DOF pan-tilt mechanism with a camera to observe the sky and identify sun or stars. By combining visual data with precise time information and orientation data from onboard sensors (IMU and magnetometer), the system estimates angular position and maintains accurate alignment.

SYSTEM COMPONENTS & TOOLS

Mechanical System

> Mechanism

2-DOF Pan-Tilt

Electronics & Embedded System

> Processing & Control Unit

Raspberry Pi 5 (8GB RAM)

Arduino Uno R3

> Vision System

OV5647 Camera (160°)

> Actuators

MG996R Servo Motors (9.4 kg-cm)

> Inertial & Navigation Sensors

Accelerometer – MPU6050 IMU

Real-Time Clock – DS3231 RTC

Magnetometer – MMC5603 3-axis

Software & Control

> Raspberry Pi OS

Python 3 (OpenCV)

HARDWARE & SENSORS

CELESTIAL NAVIGATION SYSTEM



Real-Time Clock
RTC DS3231 I2C



Magnetometer
MMC5603



Accelerometer
MPU6050 IMU



OmniVision
OV5647 160° Camera

HARDWARE & SENSORS

CELESTIAL NAVIGATION SYSTEM



Real-Time Clock
RTC DS3231 I2C



Magnetometer
MMC5603



Accelerometer
MPU6050 IMU



Omnivision
OV5647 160° Camera



Pan-Tilt Bracket



Arduino Uno R3



MG996r Servo



Raspberry Pi 5 (8Gb Ram)

5-DOF Robotic Manipulator



DESCRIPTION

Designed and developed a 5-DOF robotic arm using 3D-printed components and servo motors, focusing on precision motion control and modular mechanical design. The system integrates an Arduino-based control architecture to execute coordinated joint movements, enabling pick-and-place operations and trajectory following.

SYSTEM COMPONENTS & TOOLS

Mechanical Assembly	
> Structure	3D Printed
> Bearings	6705 Axial 51109 Thrust

Electronics & Embedded System

- > Processing & Control Unit
- > Actuators

Arduino Uno

Feetech STS3215 Servo
NEMA 17 Stepper Motor

Software & Control

- > Arduino

C/C++

HARDWARE & SENSORS

Robotic Manipulator



6705 Bearing



Nema 17 Stepper Motor



Feetech ST3215 Servo



Arduino Uno R4

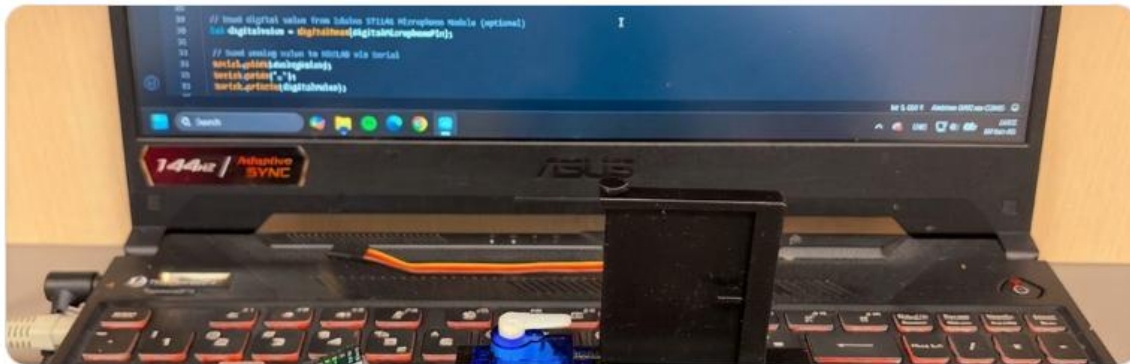


51109 Bearing



MG996r Servo

Voice-Activated Smart Door



DESCRIPTION

Developed a smart dog door that opens upon recognizing the dog's voice. Implemented voice signal processing in MATLAB and Arduino-based control, with a servo motor actuating the door.

SYSTEM COMPONENTS & TOOLS

Mechanical System

> Structure

3D Printed

Electronics & Embedded System

> Processing & Control Unit

Arduino Uno R3

> Actuators

SG-90 Servo

Electronics & Embedded System

- › Processing & Control Unit
- › Actuators
- › Sensors

Arduino Uno R3

SG-90 Servo

Microphone – Iduino ST1146

Sound Sensor – Grove LM386

Indicator – 5mm LED Diodes

Software & Control

- › Signal Processing & Identification
- › Arduino

MATLAB

C/C++

HARDWARE & SENSORS

Voice-Activated Smart Door



Grove - LM386 Sound Sensor



Microphone - Iduino ST1146

